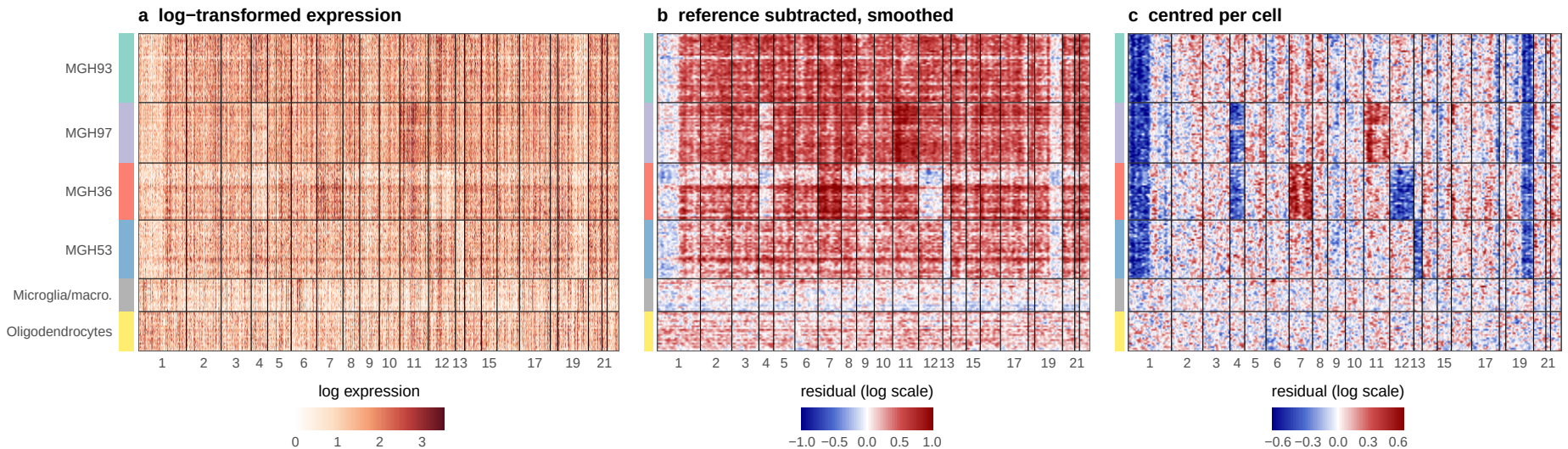
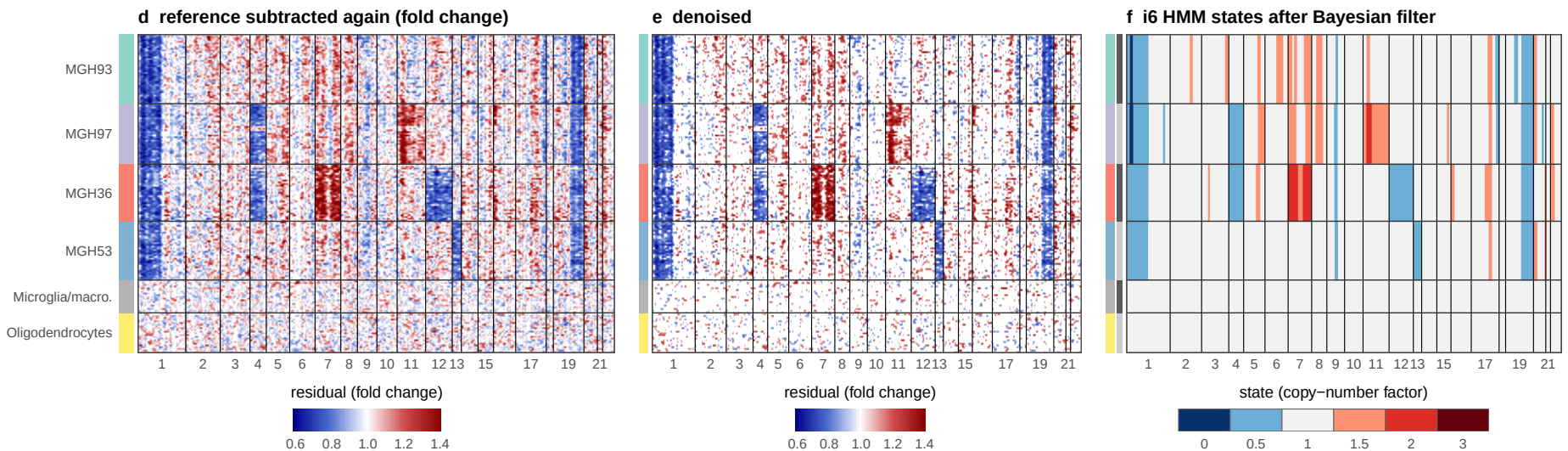

h−spike: cells simulated from the reference, processed identically
state means, sd
Counts, cell annotations, gene positions
a
Filter genes; depth−normalise; log transform
b
Subtract reference mean; smooth along chromosome
c
Centre each cell
d
Subtract reference again; to fold change
Leiden subclusters
i6 HMM per subcluster
f
Bayesian filter: drop CNVs with P(normal) > 0.5
e
Denoise (display only)


a log−transformed expression
b reference subtracted, smoothed
c centred per cell
MGH93
MGH97
MGH36
MGH53
Microglia/macro.
Oligodendrocytes
1 2 3 4 5 6 7 8 9 10 11 12 13 15 17 19 21
log expression
0 1 2 3
residual (log scale)
−1.0 −0.5 0.0 0.5 1.0
residual (log scale)
−0.6 −0.3 0.0 0.3 0.6


d reference subtracted again (fold change)
e denoised
f i6 HMM states after Bayesian filter
MGH93
MGH97
MGH36
MGH53
Microglia/macro.
Oligodendrocytes
1 2 3 4 5 6 7 8 9 10 11 12 13 15 17 19 21
residual (fold change)
0.6 0.8 1.0 1.2 1.4
residual (fold change)
0.6 0.8 1.0 1.2 1.4
state (copy−number factor)
0 0.5 1 1.5 2 3